

PRÁCTICO DE ANÁLISIS MATEMÁTICO I

LÍMITES:

Para resolver ejercicios de Límites, debemos reemplazar "x" por el valor hacia el cual tiende la misma:

$$1) \lim_{x \rightarrow 1} \frac{3x - 1}{x + 2} = \left(\lim_{x \rightarrow 1} \right) \frac{3 \cdot 1 - 1}{1 + 2} = \frac{2}{3} \rightarrow \text{Verdadero valor.}$$

↓
una vez que se reemplaza a "x"
no se pone más $\lim_{x \rightarrow 1}$

Pero en límites pueden aparecer casos de indeterminación de los mismos, que se deberán resolver "levantando" esa indeterminación... Existen 7 casos de indeterminación de límites: $\frac{0}{0}$; $\frac{\infty}{\infty}$; $0 \cdot \infty$; $\infty - \infty$; 0^0 ; ∞^0 y 1^∞ .

Cuando un límite da como resultado uno de estos 7 casos, se debe resolver con artificios de cálculo matemático...

$$2) \lim_{x \rightarrow 3} \frac{x^2 - 3x}{x - 3} = \frac{0}{0}$$

Se puede sacar factor común en el numerador:

$$\lim_{x \rightarrow 3} \frac{x(x-3)}{x-3} = \lim_{x \rightarrow 3} x = 3 \text{ (verdadero valor)}$$

$$3) \lim_{x \rightarrow -2} \frac{2x + x^2}{x + 2} = -2$$

$$4) \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 2x + 1} = \frac{0}{0} \text{ hallamos las raíces del denominador y factorizamos.}$$

$$\lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)^2} = \lim_{x \rightarrow 1} \frac{x+1}{x-1} = \frac{1+1}{1-1} = \frac{2}{0} = \infty$$

$$5) \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = 3$$

$$6) \lim_{x \rightarrow 4} \frac{x^2 - 5x + 4}{x^2 - 2x - 8} = \lim_{x \rightarrow 4} \frac{(x-4)(x-1)}{(x-4)(x+2)} = \frac{4-1}{4+2} = \frac{3}{6} = \frac{1}{2}$$

$$7) \lim_{x \rightarrow -5} \frac{x+5}{x^2-25} = -\frac{1}{10}$$

$$8) \lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x^3 - 3x^2 + 2x - 6} = \lim_{x \rightarrow 3} \frac{(x-1)(x-3)}{(x^2+2)(x-3)} = \frac{3-1}{9+2} = \frac{2}{11}$$

$$9) \lim_{x \rightarrow \infty} \frac{x^3 + 8}{x - 2} = \infty$$

$$10) \lim_{x \rightarrow -\frac{1}{5}} \frac{5x+1}{5x^2-14x-3} = \frac{25}{16}$$

$$11) \lim_{x \rightarrow -3} \frac{x^3 + 27}{3+x} = 27$$

$$12) \lim_{x \rightarrow 5} \frac{x-5}{\sqrt{x}-\sqrt{5}} = \lim_{x \rightarrow 5} \frac{(x-5)(\sqrt{x}+\sqrt{5})}{(\sqrt{x}-\sqrt{5})(\sqrt{x}+\sqrt{5})} = \lim_{x \rightarrow 5} \frac{(x-5)(\sqrt{x}+\sqrt{5})}{(x-5)} = \sqrt{5} + \sqrt{5} = 2\sqrt{5}$$

$$13) \lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1} = 2$$

$$14) \lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x} = \frac{1}{2}$$

$$15) \lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2} = \lim_{x \rightarrow 2} \frac{(\sqrt{x} - \sqrt{2})(\sqrt{x} + \sqrt{2})}{(x - 2)(\sqrt{x} + \sqrt{2})} = \lim_{x \rightarrow 2} \frac{1}{(\sqrt{x} + \sqrt{2})} = \frac{1}{\sqrt{2} + \sqrt{2}} = \frac{1}{2\sqrt{2}} = \frac{1}{2\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2(\sqrt{2})^2} = \frac{\sqrt{2}}{4}$$

$$\begin{aligned}
 16) \lim_{x \rightarrow 0} \frac{2x^2 \cos^2 x - 2x^2}{x \cos x - x} &= \lim_{x \rightarrow 0} \frac{\cancel{x} (2x \cos^2 x - 2x)}{\cancel{x} (\cos x - 1)} \\
 &= \lim_{x \rightarrow 0} \frac{2x (\cos^2 x - 1)}{\cos x - 1} = \lim_{x \rightarrow 0} \frac{2x (\cancel{\cos x - 1}) (\cos x + 1)}{\cancel{(\cos x - 1)}} \\
 &= \frac{2 \cdot 0 (\cos 0 + 1)}{1} = 0
 \end{aligned}$$

$$\begin{aligned}
 17) \lim_{x \rightarrow 7} \frac{2 - \sqrt{x-3}}{x^2 - 49} &= \lim_{x \rightarrow 7} \frac{(2 - \sqrt{x-3})(2 + \sqrt{x-3})}{x^2 - 49 (2 + \sqrt{x-3})} \\
 &= \lim_{x \rightarrow 7} \frac{4 - x + 3}{(x-7)(x+7)(2 + \sqrt{x-3})} = \lim_{x \rightarrow 7} \frac{(7-x)}{(x-7)(x+7)(2 + \sqrt{x-3})} \left\{ \begin{array}{l} 7-x = -(-7+x) \\ = -(x-7) \end{array} \right. \\
 &= \lim_{x \rightarrow 7} \frac{-\cancel{(x-7)}}{\cancel{(x-7)}(x+7)(2+2)} = -\frac{1}{6(7+7)} = -\frac{1}{56}
 \end{aligned}$$

$$18) \lim_{x \rightarrow 0} \frac{\sqrt{x+a} - \sqrt{a}}{x} = \frac{\sqrt{a}}{2a}$$

$$19) \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x} = 1$$

$$20) \lim_{x \rightarrow \infty} \sqrt{x+5} - \sqrt{x} = 0$$

$$21) \lim_{x \rightarrow \infty} \sqrt{x^2+1} - \sqrt{x} = \infty$$

$$\begin{aligned}
 22) \lim_{x \rightarrow \infty} \frac{\sqrt{x^2+3}}{x+1} &= \lim_{x \rightarrow \infty} \frac{\sqrt{x^2(1+\frac{3}{x^2})}}{x+1} = \lim_{x \rightarrow \infty} \frac{x \sqrt{1+\frac{3}{x^2}}}{x+1} \\
 &= \lim_{x \rightarrow \infty} \frac{\cancel{x} \sqrt{1+\frac{3}{x^2}}}{\cancel{x} (1+\frac{1}{x})} = \frac{\sqrt{1+\frac{3}{\infty}}}{1+\frac{1}{\infty}} = \frac{\sqrt{1+0}}{1+0} = \frac{1}{1} = 1
 \end{aligned}$$

$$23) \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x} - \sqrt{x+2}} = -\infty$$

$$24) \lim_{x \rightarrow 0} \frac{1 - 2^{1/x}}{1 + 2^{1/x}} = \lim_{x \rightarrow 0} \frac{\frac{1}{2^{1/x}} - \frac{2^{1/x}}{2^{1/x}}}{\frac{1}{2^{1/x}} + \frac{2^{1/x}}{2^{1/x}}} = \lim_{x \rightarrow 0} \frac{\frac{1}{2^{1/x}} - 1}{\frac{1}{2^{1/x}} + 1} =$$

$$= \frac{\frac{1}{2^\infty} - 1}{\frac{1}{2^\infty} + 1} = \frac{0 - 1}{0 + 1} = -1$$

$$25) \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{1 - \tan x} = -\frac{\sqrt{2}}{2}$$

$$26) \lim_{x \rightarrow \frac{3\pi}{2}} (\sin x + 1) \tan^2 x = \lim_{x \rightarrow \frac{3\pi}{2}} (\sin x + 1) \frac{\sin^2 x}{\cos^2 x} = \begin{cases} \sin^2 x + \cos^2 x = 1 \\ \cos^2 x = 1 - \sin^2 x \end{cases}$$

$$= \lim_{x \rightarrow \frac{3\pi}{2}} (\sin x + 1) \frac{\sin^2 x}{(1 - \sin^2 x)} = \lim_{x \rightarrow \frac{3\pi}{2}} \frac{(\sin x + 1) \sin^2 x}{(1 - \sin x)(1 + \sin x)} =$$

$$= \frac{\sin^2 \frac{3\pi}{2}}{1 - \sin \frac{3\pi}{2}} = \frac{(-1)^2}{1 - (-1)} = \frac{1}{2}$$

$$27) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot^2 x}{\sin x - 1} = 2$$

$$28) \lim_{x \rightarrow \pi} \frac{\cos \frac{x}{2}}{(\cos x + 1)^2} = 0$$

RESOLUCIÓN DE LOS EJERCICIOS DE LÍMITES

3) $\lim_{x \rightarrow -2} \frac{2x + x^2}{x + 2} =$ SACO FACTOR COMÚN EN EL NUMERADOR

$$= \lim_{x \rightarrow -2} \frac{x \overset{1}{(2+x)}}{\underset{1}{x+2}} = \lim_{x \rightarrow -2} x = \boxed{-2} \checkmark$$

5) $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} =$ FACTOREANDO $= \lim_{x \rightarrow 2} \frac{(x-2)(x+1)}{(x-2)} =$

$$= 2 + 1 = \boxed{3} \checkmark$$

7) $\lim_{x \rightarrow -5} \frac{x + 5}{x^2 - 25} =$ EL DENOMINADOR ES UNA DIFERENCIA DE CUADRADOS.

$$= \lim_{x \rightarrow -5} \frac{\overset{1}{x+5}}{\underset{1}{(x-5)(x+5)}} = \lim_{x \rightarrow -5} \frac{1}{x-5} = \frac{1}{-5-5} = \boxed{-\frac{1}{10}} \checkmark$$

9) $\lim_{x \rightarrow \infty} \frac{x^3}{x^3 + 8} =$ DIVIDO NUMERADOR Y DENOMINADOR POR x^3

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^3}{x^3} + \frac{8}{x^3}}{\frac{x^3}{x^3} + \frac{8}{x^3}} = \lim_{x \rightarrow \infty} \frac{1 + \frac{8}{x^3}}{1 + \frac{8}{x^3}} =$$
$$\lim_{x \rightarrow \infty} \frac{\frac{x}{x^3} - \frac{2}{x^3}}{\frac{x}{x^3} - \frac{2}{x^3}} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2} - \frac{2}{x^3}}{\frac{1}{x^2} - \frac{2}{x^3}} =$$

$$= \frac{1 + \frac{8}{\infty}}{\frac{1}{\infty} - \frac{2}{\infty}} = \frac{1 + 0}{0 - 0} = \frac{1}{0} = \boxed{\infty}$$

RECORDAR: $\begin{cases} \frac{1}{\infty} = 0 \\ \frac{\infty}{0} = \infty \end{cases}$

$$10) \lim_{x \rightarrow -\frac{1}{5}} \frac{5x+1}{5x^2-14x-3} = \text{FACTOREO EL DENOMINADOR} = \lim_{x \rightarrow -\frac{1}{5}} \frac{5x+1}{\cancel{5}\left(x+\frac{1}{5}\right)(x-3)} =$$

$$= \text{SACO FACTOR COMÚN "5" EN EL NUMERADOR} = \lim_{x \rightarrow -\frac{1}{5}} \frac{\cancel{5}\left(x+\frac{1}{5}\right)}{\cancel{5}\left(x+\frac{1}{5}\right)(x-3)} =$$

$$= \lim_{x \rightarrow -\frac{1}{5}} \frac{1}{x-3} = \frac{1}{-\frac{1}{5}-3} = \frac{1}{-\frac{16}{5}} = \boxed{-\frac{5}{16}} \checkmark_v$$

$$11) \lim_{x \rightarrow -3} \frac{x^3+27}{3+x} = \text{FACTOREAMOS EL NUMERADOR} =$$

$$= \lim_{x \rightarrow -3} \frac{\cancel{(x+3)}(x^2-3x+9)}{\cancel{(3+x)}} = \text{REEMPLAZO "x" POR } (-3) = (-3)^2 - 3(-3) + 9 =$$

$$= \boxed{27} \checkmark_v$$

$$13) \lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1} = \text{MULTIPLICO NUMERADOR Y DENOMINADOR POR LA CONJUGADA DEL DENOMINADOR:}$$

$$\lim_{x \rightarrow 1} \frac{(x-1)(\sqrt{x}+1)}{(\sqrt{x}-1)(\sqrt{x}+1)} = \text{EL DENOMINADOR ES UNA DIFERENCIA DE CUADRADOS}$$

$$\lim_{x \rightarrow 1} \frac{\cancel{(x-1)}(\sqrt{x}+1)}{\cancel{(x-1)}} = \lim_{x \rightarrow 1} \sqrt{x}+1 = \text{REEMPLAZO "x" POR } (1)$$

$$= \sqrt{1}+1 = \boxed{2} \checkmark_v$$

14) $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x} =$ MULTIPLICO NUMERADOR Y DENOMINADOR POR LA CONJUGADA DEL NUMERADOR:

$= \lim_{x \rightarrow 0} \frac{(\sqrt{x+1} - 1)(\sqrt{x+1} + 1)}{x (\sqrt{x+1} + 1)} =$ EN EL NUMERADOR TENEMOS UNA DIFERENCIA DE CUADRADOS.

$= \lim_{x \rightarrow 0} \frac{\overset{1}{\cancel{x}}}{\underset{1}{\cancel{x}} (\sqrt{x+1} + 1)} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+1} + 1} =$

$=$ REEMPLAZO "x" POR (0) $= \frac{1}{\sqrt{0+1} + 1} = \frac{1}{1+1} = \boxed{\frac{1}{2}}$ V.V.

RESOLUCIÓN DE EJERCICIOS DE LÍMITES.

18) $\lim_{x \rightarrow 0} \frac{\sqrt{x+a} - \sqrt{a}}{x}$ = MULTIPLICAMOS NUM. Y DEN. POR LA CONJUGADA DEL NUMERADOR.

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{(\sqrt{x+a} - \sqrt{a})(\sqrt{x+a} + \sqrt{a})}{x(\sqrt{x+a} + \sqrt{a})} = \lim_{x \rightarrow 0} \frac{(\sqrt{x+a})^2 - (\sqrt{a})^2}{x(\sqrt{x+a} + \sqrt{a})} \\ &= \lim_{x \rightarrow 0} \frac{x + a - a}{x(\sqrt{x+a} + \sqrt{a})} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+a} + \sqrt{a})} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+a} + \sqrt{a}} = \frac{1}{2\sqrt{a}} = \boxed{\frac{\sqrt{a}}{2a}} \quad V.V \end{aligned}$$

RACIONALIZAMOS: $\frac{1}{2\sqrt{a}} \cdot \frac{\sqrt{a}}{\sqrt{a}} = \frac{\sqrt{a}}{2(\sqrt{a})^2} = \frac{\sqrt{a}}{2a}$

19) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$ = $\lim_{x \rightarrow 0} \frac{(\sqrt{1+x} - \sqrt{1-x})(\sqrt{1+x} + \sqrt{1-x})}{x(\sqrt{1+x} + \sqrt{1-x})} =$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{1+x})^2 - (\sqrt{1-x})^2}{x(\sqrt{1+x} + \sqrt{1-x})} = \lim_{x \rightarrow 0} \frac{1+x - (1-x)}{x(\sqrt{1+x} + \sqrt{1-x})} =$$

$$= \lim_{x \rightarrow 0} \frac{1+x - 1+x}{x(\sqrt{1+x} + \sqrt{1-x})} = \lim_{x \rightarrow 0} \frac{2x}{x(\sqrt{1+x} + \sqrt{1-x})} =$$

$$= \lim_{x \rightarrow 0} \frac{2}{(\sqrt{1+x} + \sqrt{1-x})} = \frac{2}{\sqrt{1} + \sqrt{1}} = \frac{2}{1+1} = \frac{2}{2} = \boxed{1}$$

$$20) \lim_{x \rightarrow \infty} \sqrt{x+5} - \sqrt{x} = \lim_{x \rightarrow \infty} \frac{(\sqrt{x+5} - \sqrt{x})(\sqrt{x+5} + \sqrt{x})}{\sqrt{x+5} + \sqrt{x}} =$$

$$\lim_{x \rightarrow \infty} \frac{(\sqrt{x+5})^{\cancel{2}} - (\sqrt{x})^{\cancel{2}}}{\sqrt{x+5} + \sqrt{x}} = \lim_{x \rightarrow \infty} \frac{\cancel{x} + 5 - \cancel{x}}{\sqrt{x+5} + \sqrt{x}} = \lim_{x \rightarrow \infty} \frac{5}{\sqrt{x+5} + \sqrt{x}} =$$

$$= \frac{5}{\sqrt{\infty+5} + \sqrt{\infty}} = \frac{5}{\infty + \infty} = \frac{5}{\infty} = \boxed{0} \swarrow V_v.$$

$$21) \lim_{x \rightarrow \infty} \sqrt{x^2+1} - \sqrt{x} = \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2+1} - \sqrt{x})(\sqrt{x^2+1} + \sqrt{x})}{(\sqrt{x^2+1} + \sqrt{x})} =$$

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2+1})^{\cancel{2}} - (\sqrt{x})^{\cancel{2}}}{\sqrt{x^2+1} + \sqrt{x}} = \lim_{x \rightarrow \infty} \frac{x^2 - x + 1}{\sqrt{x^2+1} + \sqrt{x}} = \text{SACAMOS FACTOR COMÚN "X" EN EL NUMER. Y EL DENOM.}$$

$$\text{NUMERADOR: } x^2 - x + 1 = x(x - 1 + \frac{1}{x})$$

$$\text{DENOMINADOR: } \sqrt{x^2(1 + \frac{1}{x^2})} + \sqrt{x^2(\frac{1}{x})} = x\sqrt{1 + \frac{1}{x^2}} + x\sqrt{\frac{1}{x}} = x(\sqrt{1 + \frac{1}{x^2}} + \sqrt{\frac{1}{x}})$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x} * (x - 1 + \frac{1}{x})}{\cancel{x} * (\sqrt{1 + \frac{1}{x^2}} + \sqrt{\frac{1}{x}})} = \frac{\infty - 1 + \frac{1}{\infty}}{\sqrt{1 + \frac{1}{\infty}} + \sqrt{\frac{1}{\infty}}} = \frac{\infty}{1} = \boxed{\infty} \swarrow V_v.$$

$$\begin{aligned}
 23) \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x} - \sqrt{x+2}} &= \lim_{x \rightarrow \infty} \frac{x (\sqrt{x} + \sqrt{x+2})}{(\sqrt{x} - \sqrt{x+2})(\sqrt{x} + \sqrt{x+2})} = \\
 \lim_{x \rightarrow \infty} \frac{x (\sqrt{x} + \sqrt{x+2})}{(\cancel{\sqrt{x}})^2 - (\cancel{\sqrt{x+2}})^2} &= \lim_{x \rightarrow \infty} \frac{x (\sqrt{x} + \sqrt{x+2})}{x - (x+2)} = \\
 = \lim_{x \rightarrow \infty} \frac{x (\sqrt{x} + \sqrt{x+2})}{\cancel{x} - \cancel{x} - 2} &= \frac{\infty}{-2} = \boxed{-\infty} \text{ V.V.}
 \end{aligned}$$

$$\begin{aligned}
 25) \lim_{x \rightarrow \frac{\pi}{4}} \frac{\text{sen } x - \cos x}{1 - \tan x} &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\text{sen } x - \cos x}{1 - \frac{\text{sen } x}{\cos x}} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\text{sen } x - \cos x}{\frac{\cos x - \text{sen } x}{\cos x}} = \\
 = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x (\text{sen } x - \cos x)}{\cos x - \text{sen } x} &= \begin{cases} \text{SACAMOS FACTOR COMÚN } (-1) \\ \text{EN } \cos x - \text{sen } x : \\ \cos x - \text{sen } x = -(-\cos x + \text{sen } x) \\ = -(\text{sen } x - \cos x) \end{cases} \\
 = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x (\text{sen } x - \cos x)}{- (\text{sen } x - \cos x)} &= \lim_{x \rightarrow \frac{\pi}{4}} -\cos x = \boxed{-\frac{\sqrt{2}}{2}} \text{ V.V.}
 \end{aligned}$$

$$27) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot^2 x}{\text{sen } x - 1} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{\cos^2 x}{\text{sen}^2 x}}{\text{sen } x - 1} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x}{\text{sen}^2 x (\text{sen } x - 1)} =$$

POR TRIGONOMETRÍA: $\text{sen}^2 x + \cos^2 x = 1 \rightarrow \cos^2 x = 1 - \text{sen}^2 x$

$$\begin{aligned}
 = \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \text{sen}^2 x}{\text{sen}^2 x (\text{sen } x - 1)} &= \begin{aligned} &1 - \text{sen}^2 x = \text{DIFERENCIA DE CUADRADOS} \\ &= (1 - \text{sen } x)(1 + \text{sen } x) \end{aligned}
 \end{aligned}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{(1 - \tan x)(1 + \tan x)}{\tan^2 x (\tan x - 1)} = \text{SAO FACTOR COMUN}(-1) \text{ EN } \tan x - 1 \text{ o } \tan x - 1 \text{ o}$$

$$\tan x - 1 = -(-\tan x + 1)$$

$$= -(1 - \tan x)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cancel{(1 - \tan x)} (1 + \tan x)}{-\tan^2 x \cancel{(1 - \tan x)}} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 + \tan x}{-\tan^2 x} =$$

$$= -\frac{1 + 1}{1} = \boxed{-2} \text{ V.V.}$$

$$28) \lim_{x \rightarrow \pi} \frac{\cos \frac{\pi}{2}}{(\cos x + 1)^2} = \lim_{x \rightarrow \pi} \frac{\cos \frac{\pi}{2}}{(\cos^2 x + 2 \cos x \cdot 1 + 1^2)} = \begin{cases} \tan^2 x + \cos^2 x = 1 \\ \therefore \cos^2 x = 1 - \tan^2 x \\ \therefore \cos x = \sqrt{1 - \tan^2 x} \end{cases}$$

$$= \lim_{x \rightarrow \pi} \frac{\cos \frac{\pi}{2}}{(1 - \tan^2 x) + 2 \sqrt{1 - \tan^2 x} + 1} = \frac{0}{1 - 0 + 2 \cdot 0 + 1} = \frac{0}{2} = \boxed{0} \text{ V.V.}$$